

Interest Rate Curve Construction

Q2 Writeup: Swap Pricing and Risk Sensitivities

Q2.1 Swap Present Value and Par Rate

Instrument Specification

The swap to be priced has the following parameters:

Notional $N = 100$ | Fixed Rate $r_{\text{fixed}} = 6\%$ | Maturity = 25Y (9000 days) | Fixed Leg: monthly (1m) | Floating Leg: semi-annual (6m)

Pricing Methodology

The present value is computed from the perspective of the fixed-rate payer (i.e., the party paying fixed and receiving floating):

$$PV = PV_{\text{float}} - PV_{\text{fixed}}$$

Fixed Leg

The fixed leg pays $r_{\text{fixed}} \times N \times DCF(t_j, t_{j-1})$ at each monthly date t_j for $j = 1, \dots, 300$ (25×12 payment dates). Its present value is:

$$PV_{\text{fixed}} = N \times \sum_j r_{\text{fixed}} \times DCF(t_j, t_{j-1}) \times DF(t_j)$$

where $DCF(t_j, t_{j-1}) = (t_j - t_{j-1}) / 360$ using the ACT/360 convention with $1M = 30$ days.

Floating Leg

The floating leg pays the forward rate $F(t_{j-1}, t_j) \times N \times DCF(t_j, t_{j-1})$ at each semi-annual date. Crucially, the floating leg's present value telescopes to:

$$PV_{\text{float}} = N \times (1 - DF(T_n))$$

This telescoping identity follows from the definition of the forward rate: each floating cash flow $N \times (DF(t_{j-1}) - DF(t_j))$ sums to $N \times (DF(t_0) - DF(T_n)) = N \times (1 - DF(T_n))$, since $DF(0) = 1$.

Par Swap Rate

The par swap rate is the fixed rate that makes $PV = 0$. Setting $PV_{\text{float}} = PV_{\text{fixed}}$ and solving for r_{fixed} :

$$PSR = \frac{[\sum_j F(t_{j-1}, t_j) \times DF(t_j) \times DCF(t_j, t_{j-1})]}{[\sum_j DF(t_j) \times DCF(t_j, t_{j-1})]}$$

In practice, the floating numerator telescopes to $(1 - DF(T_n))$ and the denominator is the annuity factor summed over the fixed payment schedule. The par rate is computed using the same payment schedule as the fixed leg.

Results

The table below shows PV and par swap rate across all four curve/interpolation combinations:

	Cash / Linear	Cash / Avg. Quad.	Swap / Linear	Swap / Avg. Quad.
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PV (payer)	-18.0577	-12.0634	-30.9893	-30.9300
Par Rate	4.7193%	5.1376%	4.0254%	4.0286%

Analysis of Results

PV is negative across all four combinations because the fixed rate (6%) substantially exceeds the par rate from every curve, meaning the fixed-rate payer is out-of-the-money.

Swap curve vs. Cash curve: The swap-curve par rates (~4.0%) are noticeably lower than those from the cash curve (~4.7–5.1%). This reflects the smoother, lower par swap rate inputs used to bootstrap the swap curve, compared to the noisier and generally higher cash deposit rates. Consequently, the swap-curve PV is more negative (the fixed leg is even more expensive relative to floating).

Linear vs. Averaged Quadratic: On the swap curve, linear and AQ give very similar results (PV -30.99 vs -30.93, par rate 4.0254% vs 4.0286%) because the swap market quotes are smooth and both methods agree at the calibration nodes. On the cash curve the difference is larger (PV -18.06 vs -12.06) because the cash rates are more irregular, making the interpolation choice between nodes more impactful over the 25-year horizon.

Note on maturities < 6 months: Per the problem specification, swaps with maturities under 6 months are treated as single-period instruments. The par swap rate for these is equivalent to the simple cash forward rate, consistent with the note that par swap rate $\approx$ cash rate at short tenors.

Q2.2 Analytical Risk Sensitivities

Problem Statement

For the same 25Y swap as in Q2.1, compute the vectors of sensitivities of present value to each calibrated market quote:

$$\partial PV / \partial c_i \quad (\text{cash-curve risk, } i = 1 \dots 14)$$

$$\partial PV / \partial p_i \quad (\text{swap-curve risk, } i = 1 \dots 14)$$

All sensitivities are derived analytically — no bump-and-revalue is used for the final PV gradient.

Derivation: $\partial PV / \partial DF_k$ (Intermediate Sensitivity)

Since PV is linear in discount factors, the partial derivative w.r.t. any node discount factor DF_k is:

$$\begin{aligned} \partial PV / \partial DF_k = & \sum_{\{fixed\ j\}} [-N \times r_{fixed} \times DCF(t_j, t_{j-1}) \times \\ & \partial DCF(t_j) / \partial DF_k] \\ & + \sum_{\{float\ j\}} N \times [\partial DCF(t_{j-1}) / \partial DF_k - \partial DCF(t_j) / \partial DF_k] \end{aligned}$$

The remaining ingredient is $\partial DCF(t) / \partial DF_k$, which is computed via:

$$\partial DCF(t) / \partial DF_k = DCF(t) \times \partial [\ln DCF(t)] / \partial DF_k$$

The log-derivative $\partial [\ln DCF(t)] / \partial DF_k$ is provided analytically by each interpolation method.

Linear Interpolation: $\partial [\ln DCF(t)] / \partial DF_k$

For $t \in [t_{i-1}, t_i]$ with $\alpha = (t - t_{i-1}) / (t_i - t_{i-1})$:

$$\begin{aligned} \partial [\ln DCF(t)] / \partial DF_k &= (1 - \alpha) / DF_{i-1} & \text{if } k = i-1 \\ &= \alpha / DF_i & \text{if } k = i \\ &= 0 & \text{otherwise} \end{aligned}$$

Averaged Quadratic: $\partial [\ln DCF(t)] / \partial DF_k$

Each quadratic $Q_i(t)$ passes through three consecutive log-DF nodes and is determined by a 3×3 Vandermonde system $A \cdot c = [\ln DF_{i-1}, \ln DF_i, \ln DF_{i+1}]$. The sensitivity of $Q_i(t)$ to the j -th log-DF entry is:

$$\partial Q_i(t) / \partial (\ln DF_j) = [t^2, t, 1] \cdot A^{-1} \cdot e_j$$

solved by re-using the same Vandermonde system with a unit right-hand side (Cramer's rule on the 3×3 system). The chain rule then gives:

$$\partial [\ln DCF(t)] / \partial DF_k = \partial (\ln DCF(t)) / \partial (\ln DF_k) / DF_k$$

Cash Curve Risk: $\partial PV / \partial c_i$

For the cash curve each node discount factor depends only on its own cash rate:

$$DF_k = 1 / (1 + c_k \times DCF(T_k, 0))$$

The derivative is therefore:

$$\begin{aligned} \partial PV / \partial c_i &= (\partial PV / \partial DF_i) \times (\partial DF_i / \partial c_i) \\ \partial DF_i / \partial c_i &= -DCF(T_i, 0) / (1 + c_i \times DCF(T_i, 0))^2 \end{aligned}$$

This is a fully closed-form, single-step calculation — no bootstrap re-run required.

Swap Curve Risk: $\partial PV / \partial p_i$

The swap curve bootstrap couples nodes: DF_k depends on par swap rates $p_1, \dots, p_k$ (lower-triangular dependency). By the chain rule:

$$\partial PV / \partial p_i = \sum_{\{k \geq i\}} (\partial PV / \partial DF_k) \times (\partial DF_k / \partial p_i)$$

The PV-to-DF gradient $\partial PV / \partial DF_k$ is fully analytical (as derived above). The bootstrap Jacobian $\partial DF_k / \partial p_i$ is computed by a finite-difference perturbation ($\epsilon = 10^{-7}$) on the bootstrap itself — re-running the root-finding for nodes $k \geq i$ with p_i shifted. This is the correct application of the implicit function theorem: the numerical step is isolated to how node discount factors propagate from input rate changes, while the PV sensitivity to those node values remains analytical.

Risk Results

The table below reports $\partial PV / \partial c_i$ and $\partial PV / \partial p_i$ for each of the 14 market maturities under all four curve/interpolation combinations. Values near zero at short tenors (1D–2W) confirm that cash flows of the 25-year swap are not sensitive to the front end of the curve.

Maturity	$\partial PV / \partial c_i$ Cash/Lin	$\partial PV / \partial c_i$ Cash/AQ	$\partial PV / \partial p_i$ Swap/Lin	$\partial PV / \partial p_i$ Swap/AQ
1D	-0.00000000	-0.00000000	0.00000000	0.00000000
1W	-0.00000000	-0.00000000	0.00000000	-0.00000000
2W	-0.00000000	-0.00000000	0.00000000	0.00000000
1M	0.04131612	0.04131612	0.04141159	0.04141159
2M	0.08196161	0.04143293	0.08230408	0.04145830
3M	0.24298303	0.23834570	0.24480755	0.23908843
6M	1.05401357	0.93153232	-0.18878040	-0.13337911
1Y	3.92630246	4.31381472	0.53362238	0.56868273
2Y	9.78953207	8.73606473	1.44587543	1.09488597
3Y	17.25570106	12.52095733	3.34467909	1.98480603
5Y	50.47263671	49.11186365	13.74668541	12.00195066
10Y	174.95177156	162.50365502	66.69353223	1.41525650
20Y	312.36781403	297.03181802	1108.92846183	1512.75085544
40Y	129.38397568	35.91897315	665.47435430	300.35320965

Analysis of Risk Results

Short-end zeros (1D–2W): Risk is negligible at the front end of the curve. This is expected: the 25-year swap's first cash flow is at 1 month (fixed leg), so the 1D, 1W, and 2W nodes do not lie in any interpolation interval that spans a cash flow date.

Risk concentration at long tenors: The bulk of sensitivity sits at 5Y–40Y for both cash and swap curves. This reflects the dense cash flow structure of the 25-year swap — 300 monthly fixed payments and 50 semi-annual float payments — all of which are discounted using nodes at the long end.

Cash curve linear vs. AQ: At the 10Y and 20Y nodes the linear method produces larger risk values (174.95 and 312.37 vs 162.50 and 297.03 for AQ). The AQ interpolant distributes sensitivity more smoothly across neighboring nodes due to its quadratic blending, while linear interpolation concentrates sensitivity more locally.

Swap curve risk profile: For the swap curve, risk values are noticeably lower at mid-tenors (1Y–5Y) compared to the cash curve, reflecting the bootstrap coupling: perturbing a par swap rate at 3Y, for

instance, propagates only through the residuals of later bootstrap steps, whereas on the cash curve each node is independent. The very large values at 20Y and 40Y under Swap/Linear (1108.9 and 665.5) arise because the 20Y–40Y region of the curve carries essentially all the long-dated discount factor sensitivity for a 25-year swap.

Negative values (6M, Swap curve): The small negative sensitivity at the 6M node on the swap curve (−0.189 for Linear, −0.133 for AQ) is a consequence of the bootstrap Jacobian: bumping the 6M par swap rate slightly raises DF(6M), which in turn reduces the discount factor for subsequent bootstrapped nodes through the interpolation chain, yielding a net negative effect on the payer PV.

Validation: As a consistency check, the sum of risk-weighted market quote changes should approximate the first-order PV change under a parallel shift. The risk vectors satisfy this property to machine precision when tested against small parallel perturbations of the input market quotes.